

HUJI Research Monitor

Healthcare · Weekly Digest · Week 28 · July 06, 2026

EXECUTIVE SUMMARY

This week's digest highlights a diverse portfolio of breakthroughs poised for significant commercial impact across biotechnology, medical devices, and diagnostics. Innovations include advanced cell and gene therapies targeting autoimmune diseases and demyelinating disorders, alongside novel precision oncology drugs and diagnostics. We also feature high-impact medical devices, from antimicrobial surface coatings to laser-based glaucoma treatments, and AI tools accelerating drug discovery. These technologies address major unmet medical needs and offer scalable solutions for chronic diseases, infection control, and enhanced diagnostics.

#1

B cell Maturation Antigen Targeted Chimeric Antigen Receptor T Cell Therapy for Refractory Systemic Lupus Erythematosus and Systemic Sclerosis.

37/50

Main Researcher: Chamutal Gur

[Immunology] [Clinical] [Synthetic Biology]

"Novel CAR T-cell therapy HBI0101 targets severe autoimmune diseases, opening a new frontier in chronic disease management."

WHY NOW

Shifting CAR T-cell technology from oncology to large, unmet autoimmune markets represents a significant commercial expansion opportunity. Early clinical data supports its promise.

COMMERCIAL ANGLE

Developing specialized CAR T-cell therapies for autoimmune indications represents a massive shift from oncology toward high-value chronic disease management markets.

<https://europepmc.org/article/MED/42098533>

#2

Pro-repair properties of a human embryonic stem cell-derived astrocyte cell therapy in demyelinating disorders.

36/50

Main Researcher: Tamir Ben-Hur

[Neuroscience] [Immunology] [Clinical] [Synthetic Biology]

"Scalable embryonic stem cell-derived astrocyte therapy offers an off-the-shelf solution for demyelinating neurological disorders."

WHY NOW

An off-the-shelf cell therapy addressing remyelination in MS and other disorders fills a critical gap within a high-value, chronic disease market.

COMMERCIAL ANGLE

Development of a scalable, off-the-shelf cell therapy for remyelination in Multiple Sclerosis and other demyelinating neurological disorders.

<https://pubmed.ncbi.nlm.nih.gov/42276062/>

#3

33/50

The effect of a proprietary chlorine-stabilizing surface-coating formulation on healthcare environment disinfection in multidrug-resistant bacteria isolation rooms.

Main Researcher: Oren Zimhony

[Medical Device] [Materials] [Clinical]

"OxiLast™ coating dramatically boosts hospital disinfection, combating multidrug-resistant bacteria on high-touch surfaces."

WHY NOW

Hospitals urgently need cost-effective, durable solutions to reduce healthcare-associated infections, especially with rising antibiotic resistance.

COMMERCIAL ANGLE

This technology offers a scalable solution for hospitals and long-term care facilities to enhance infection control protocols and reduce the transmission of multidrug-resistant organisms through high-durability antimicrobial coatings.

<https://pubmed.ncbi.nlm.nih.gov/42374536/>

#4

33/50

RNA dicing promotes the expression of an oncogenic JAK1 isoform.

Main Researcher: Yuval Malka

[Drug Discovery] [Diagnostics] [Genomics] [Immunology]

"Discovering RNA dicing profiles enables precision patient stratification for existing JAK1 inhibitors in cancer."

WHY NOW

Companion diagnostics are crucial for optimizing drug efficacy and expanding markets for approved drugs, aligning with precision oncology trends.

COMMERCIAL ANGLE

The discovery enables a new precision medicine approach by using RNA dicing profiles to stratify patients for existing JAK1 inhibitors like momelotinib.

<https://pubmed.ncbi.nlm.nih.gov/41984590/>

#5

33/50

Recombinant Human Amelogenin Protein Enhances Healing of Osteochondral Injury in a Goat Model.

Main Researcher: Amir Haze

[Medical Device] [Clinical] [Synthetic Biology] [Proteomics]

"Remelix® recombinant protein shows significant promise in regenerating cartilage and bone for osteoarthritis and joint injuries."

WHY NOW

The massive market for joint repair and osteoarthritis lacks effective regenerative treatments, positioning this biologic for high demand.

COMMERCIAL ANGLE

Development of a high-value regenerative medicine biologic for treating osteoarthritis and traumatic joint injuries in humans.

<https://pubmed.ncbi.nlm.nih.gov/41925697/>

#6

APOBEC mutational signatures predict immune checkpoint inhibitor benefit in TMB-low metastatic NSCLC independent of PD-L1 status.

32/50

Main Researcher: N Peled

[Diagnostics] [Genomics] [Immunology] [Software/AI]

"APOBEC signatures predict immunotherapy response in lung cancer, expanding eligible patients for life-saving treatments."

WHY NOW

Expanding the patient population for high-value immune checkpoint inhibitors through a novel biomarker represents a significant commercial opportunity.

COMMERCIAL ANGLE

Development of a companion diagnostic test using machine learning to identify APOBEC signatures in TMB-low lung cancer patients to expand the market for immune checkpoint inhibitors.

<https://pubmed.ncbi.nlm.nih.gov/42378718/>

#7

Targeting primary and metastatic ovarian cancer with a peptide derived from the human NAF-1/CISD2 protein.

29/50

Main Researcher: Rachel Nechushtai

[Drug Discovery] [Clinical] [Immunology]

"Novel peptide targeting NAF-1/CISD2 shows first-in-class potential for aggressive primary and metastatic ovarian cancer."

WHY NOW

High unmet need in ovarian cancer for precision therapies offers substantial market potential for this novel therapeutic candidate.

COMMERCIAL ANGLE

The peptide represents a high-potential therapeutic candidate for a first-in-class oncology drug, suitable for licensing to pharmaceutical companies focusing on precision cancer therapies or protein-protein interaction inhibitors.

<https://pubmed.ncbi.nlm.nih.gov/42259134/>

#8

Breath-Based Detection of Oral Diseases Using Sensors and Machine Learning.

28/50

Main Researcher: Yael Hour-Haddad

[Medical Device] [Diagnostics] [Software/AI]

"Non-invasive breathalyzer using AI and nanosensors offers rapid, objective diagnosis for common oral diseases."

WHY NOW

Dentists lack objective, rapid diagnostic tools for periodontal health, making a breathalyzer a highly desired and scalable solution.

COMMERCIAL ANGLE

Development of a non-invasive, rapid-screening handheld breathalyzer for dental clinics to provide objective diagnostics for periodontal health.

<https://pubmed.ncbi.nlm.nih.gov/42379259/>

#9

Targeting Temozolomide-Resistant Glioblastoma: Therapeutic Potential of Neuronal Nitric Oxide Synthase Inhibitor.

26/50

Main Researcher: Haitham Amal

[Drug Discovery] [Neuroscience] [Clinical]

"Nitric oxide synthase inhibitor overcomes temozolomide resistance in glioblastoma, extending survival for a deadly cancer."

WHY NOW

Glioblastoma's resistance to standard chemotherapy creates a critical need for adjuvant therapies that can improve patient outcomes and extend drug utility.

COMMERCIAL ANGLE

Development of a high-value adjuvant therapy to extend the clinical utility of existing standard-of-care chemotherapy for resistant brain tumors.

<https://pubmed.ncbi.nlm.nih.gov/42374654/>

#10

Strategic Template Filtering Accelerates Fragment-Based Peptide Docking.

26/50

Main Researcher: Ora Schueler-Furman

[Drug Discovery] [Software/AI] [Proteomics]

"Optimized AI algorithm accelerates peptide docking, significantly boosting efficiency for next-gen drug discovery platforms."

WHY NOW

The demand for AI-driven tools to speed up and reduce the cost of drug discovery is immense, making this algorithm highly attractive for licensing.

COMMERCIAL ANGLE

Licensing or integrating this high-efficiency docking algorithm into AI-driven drug discovery platforms to accelerate the design of peptide-based therapeutics.

<https://pubmed.ncbi.nlm.nih.gov/42299509/>

#11

Cell-to-cell variability and gain of methylation at polycomb CpG islands as a hallmark of aging.

26/50

Main Researcher: Howard Cedar

[Genomics] [Diagnostics] [Drug Discovery]

"Epigenetic markers identify cellular aging, offering new biomarkers for age tracking and targets for anti-aging therapies."

WHY NOW

Growing consumer and clinical interest in biological age tracking and anti-aging interventions creates a significant market for precise epigenetic biomarkers.

COMMERCIAL ANGLE

Development of high-precision epigenetic biomarkers for biological age tracking and novel therapeutic targets for epigenetic reprogramming to treat age-related diseases.

<https://pubmed.ncbi.nlm.nih.gov/42265112/>

#12

26/50

Safety and Efficacy of Direct Selective Laser Trabeculoplasty in Patients With Ocular Hypertension or Open Angle Glaucoma.

Main Researcher: David Zadok

[Medical Device] [Clinical]

"Direct Selective Laser Trabeculoplasty (DSL T) proves safe and effective as a first-line glaucoma treatment."

WHY NOW

Growing patient dissatisfaction with daily eye drops and demand for non-pharmacological, first-line glaucoma treatments positions DSL T for rapid adoption.

COMMERCIAL ANGLE

The findings support the commercialization and expanded clinical adoption of laser-based ophthalmic surgical devices as a first-line alternative to traditional glaucoma eye drops.

<https://europepmc.org/article/MED/42148860>